

Chapter 7

Innovation Centers and Economic Development



Kevin T. Byrne and Victor R. Clements

7.1 Introduction

This chapter provides an explanatory “how-to” guide with case-studies and real-world examples toward the development of innovation centers and their capacity to spur economic development. The information included in this chapter is not proprietary. **The University Financing Foundation (TUFF)** has learned that there are no quick fixes or uniform approaches to creating economic development from University-Industry (UI) partnerships. Instead, the chapter will help practitioners understand general principles and approaches that can be implemented in a variety of contexts.

TUFF began through activities at the Georgia Institute of Technology in 1982, when its university president, Joseph Pettit, envisioned a research enterprise that fostered a community of education and innovation. President Pettit’s mandate was to pursue high-quality buildings using a private-sector delivery model; to use a lease structure which mirrored the cost of public finance and was therefore below market; and to maximize federal funding opportunities. This vision led to an independent organization—started by Georgia Tech alums passionate about assisting the institution expand its enterprise—called The University Financing Foundation, a 501(c)(3) operating foundation. TUFF’s first engagement resulted in Georgia Tech’s J. Frank Smith Centennial Research Building, a project that followed President Pettit’s mandate and enabled the successful expansion of a vibrant research enterprise over the last 43 years.

TUFF’s model is grounded in four tenets of productive UI collaboration:

K. T. Byrne (✉) · V. R. Clements
The University Financing Foundation (TUFF), Atlanta, GA, USA
e-mail: kevin@tuff.org; vic@tuff.org

1. High-quality academic institutions (University)
2. Private investment in research and development (Industry)
3. Collaboration between University and Industry
4. The protection of intellectual property¹

If the ultimate purpose of a research park or an innovation center² is to raise the standard of living through increased commercial activity and entrepreneurship, then university and industry partners can draw from these approaches to make location-specific decisions to create mutually beneficial and positively reinforcing ecosystems. The chapter will conclude with recommendations for prospective partnerships and reflections on future trends among innovation centers nationwide.

This chapter shares lessons learned and considerations for prospective practitioners garnered from over four decades of experience, during which time TUFF has developed a nuanced understanding of success factors that can influence the trajectory of a research park. The four tenets of a research park outlined above will be specifically described through the following six criteria, which are essential to the creation, development, and sustainability of an innovation community:

1. Entrepreneurship and asset mapping
2. Academic presence
3. Academic and public support, commitment to R&D
4. Fundraising and external relations
5. Regional collaboration
6. Activation and events, placemaking in real estate

While this list is not meant to be exhaustive, the topics underscore TUFF's belief that innovation centers should strive to serve as a conduit for authentic, serendipitous human connections; the resulting free exchange of ideas spurring innovations that can ultimately transfer to and enrich the community at large.

7.2 Prior Research and Experience

There is a large, persuasive body of research on UI collaborations as engines of economic and academic activity. Research has demonstrated that innovation centers, when properly resourced and managed, drive economic growth in a region, spur innovation among industry, promote entrepreneurial activity, and contribute positively to research and academic pursuits (Dooley & Kirk, 2007). Relatedly, research confirms that these university innovation collaborations take shape very differently due to the varying characteristics among different regions of the country.

¹Intellectual property will be discussed at length in a later chapter, so our discussion of this tenet will be more limited.

²In this chapter, the terms “research parks” and “innovation centers” will be used interchangeably.

Place-based factors (e.g., cultural, economic, academic) that influence UI partnerships and the resulting infrastructure of a UI partnership differ widely. Consequently, research confirms that there is no “one-size-fits-all” model for success (Rissola et al. 2019).

Qualitative benchmarking studies have indicated that research parks are resilient and lasting vehicles for economic mobility, and a 2013 study by the Battelle Technology Partnership Practice demonstrated striking differences in job growth, startup success, and long-term retained earnings for hubs and associated companies affiliated with an innovation center or research park (Battelle, 2013). Additional research indicates the impact of regional economic policies that support innovation and are attractive locations for immigrants, who in turn experience relative increases in wealth and other socioeconomic markers as compared to other localities, while also creating a cycle of positive growth in a geography (Cullum Clark, 2022).

Regarding academic output, studies demonstrate the net positive impact of innovation centers on academic research and development, and successful technology transfer (Dooley & Kirk, 2007). As to their impact on universities, studies investigating the impact of UI partnerships on publication output among faculty have noted an increase in overall research output and an increase in patents among both small to medium enterprises and larger enterprises (Chai & Shih, 2016). Research also demonstrates that innovation districts can break down silos among universities or create positive, parallel incentives for R&D cooperation that might otherwise not exist across disciplines (Katz & Ordover, 1990).

A review of more recent literature also indicates a growing body of research on the economic impact of “innovation districts” in regions with comparatively less-established industry, fewer entrepreneurial resources, or simply smaller populations (Hutton, 2004). These studies seem to demonstrate that while there are a wide variety of factors impacting overall success, a common theme among successful districts is the adoption of a “stakeholder” approach to coalition building. Stakeholder theory was originally conceptualized in 1984 by R. Edward Freeman, a business school professor seeking to quantify business returns in the context of a network of stakeholders extending far beyond the traditional shareholder. Similarly, research parks and innovation centers that take a holistic approach to network-building appear to more efficiently leverage resources in their region and perhaps also encounter fewer administrative obstacles (Darmody & Bendis 2021). More recently, responses to the Covid-19 pandemic and associated macroeconomic policies have begun to be researched in the context of biotechnology clusters’ relative ability to grow and spur regional economic development, including the history and common characteristics of successful “clusters” (Darmody & Bendis 2021). It is important to note that research on newer UI partnerships is not exhaustive and thus far the evidence pointing to the impact of stakeholder theory is more qualitative and anecdotal.

7.3 Options, Solutions, and Impact

This section shares considerations for UI collaborations and will detail six topics that can support the creation, development, and sustainability of an innovation community.

7.3.1 *Entrepreneurship*

Entrepreneurial ecosystems are critical to successful UI partnerships and entrepreneurial activity is often a primary motivator for a variety of stakeholders. In the academic sphere, entrepreneurship is now an important counterpart to many researchers' scientific or strictly academic pursuits. Furthermore, technology transfer and successful commercialization of new technologies are some of the most productive revenue streams for a UI collaboration. Consequently, many UI partnerships dedicate significant staff time and resources towards promoting entrepreneurship and it is frequently listed as a top priority among research park staff. In the UI context, fostering entrepreneurship requires the creation of a physical and virtual space along with a corresponding set of authentic activities conducive to innovation.

Large universities have an abundance of resources that are often spread out across a variety of departments and institutes, oftentimes operating in parallel silos. These universities also have a supportive economic infrastructure in surrounding cities. In both cases, there are abundant resources and assets that may have few incentives or opportunities to coordinate their activities towards a common goal. To help better understand and align these various assets, UI partnerships should consider "asset mapping" as an important first step towards building a strong ecosystem for entrepreneurship and technology transfer. In the UI context, asset mapping is an exercise in authentic relationship building: creating a strong and diffuse network of partners and an understanding of their various strengths and priorities. Not coincidentally, entrepreneurship and asset mapping are mutually reinforcing and beneficial activities, since understanding and leveraging existing assets is frequently a core competency for entrepreneurs.

One important and relatively natural way research parks can support entrepreneurship is by serving as a convener or a translator. In understanding a broad coalition of stakeholders' interests and creating mechanisms for these stakeholders to effectively interact, a university reduces barriers to entrepreneurial activity. To help prioritize effective "asset-mapping" in a region, UI partnerships should prioritize early hiring of at least one senior staff member who is experienced and well-networked in areas like venture capital or early-stage investing. These individuals can serve as a strong, public-facing spokesperson to attract a coalition of partners and build clear understanding of a research park's capacities among its regional community. Universities can also leverage existing administrative and support

services (like staff members in a particular department or institute) to create structural support for entrepreneurship and to promote those activities to researchers and academics.

University City Science Center, in Philadelphia, has proactively worked to create a coordinated network of assets to promote entrepreneurship and technology transfer. The University City Science Center initially leveraged a nonprofit structure to create programming that incentivized relationship development among entrepreneurs and provided access to training and resources, but also focused on the relative strengths and assets native to Philadelphia. Eventually, a physical footprint was established to further concentrate entrepreneurs, academics, and industry partners in a single gathering place in the city.

The above example and many other UI partnerships underscore the role of communication in asset mapping and successful enterprises, generally. This ability to listen and learn from within the community through authentic relationship development will help an innovation center develop a practice of self-assessment in measuring its ability to meet the needs of its diverse group of stakeholders. And as a tool for culture-building, asset mapping is critical in fostering open minded innovation and collaboration, avoiding pitfalls that naturally occur among varied interests within a large university or among regional partners.

Successful innovation centers and research parks also make intentional decisions about their physical layout and programming, especially related to entrepreneurial interaction with industry. These choices should encourage seemingly disparate academic disciplines to interact and build relationships to promote collaboration and ensure the protection of IP. Decisions about the physical layout and the programmatic activities are part of the placemaking and culture-building activities that entrepreneurs and industry partners consistently report as being vital to their success and important in their initial decisions regarding colocation.

When effectively combined, clear communication and intentional programming work together to support entrepreneurship. For example, coordinated communication paired with physical density allows for more natural serendipity and collaboration in promoting unplanned interactions among scientists and entrepreneurs. While some of these efforts may be harder to measure and can appear less important, relationships and culture-building activities within a research park are an absolutely vital contributor to commercialization and technology transfer.

Challenges to creating a culture of entrepreneurship can arise when a university has less experience with technology transfer. For many years, universities viewed industry partners with skepticism and a culture of “academic purity” can occasionally persist today. This perspective naturally limits the sort of holistic collaboration across a university required to support a vibrant research park. Other times, limited or inconsistent coordination between a center for entrepreneurship and a broader research park can make collaboration less likely simply by virtue of physical distance or because programming does not effectively encourage engagement.

To counteract these forces, research parks should carefully consider their leadership staffing decisions and the structural and programmatic protections for entrepreneurship. They should work hard to build a culture that fosters creativity and

open-mindedness, the details of which are further outlined in a later section of this chapter. Focusing on the overall experience of entrepreneurs and researchers alike can help attract and retain talent of all types to a specific region. That said, these efforts require long-term investment. There will almost certainly be few quick answers. As has been well-documented, the latest generation of students and early-career workers has evolving priorities, so regions need to create aligned incentives across the entire UI partnership to keep both students and community stakeholders engaged.

Depending on local and regional industry needs, UI partnerships will need to be responsive and enterprising themselves. UI partnerships that can listen to and adapt to its stakeholders' needs will create winning programs. For example, some innovation centers partnerships have developed job training programs and interfaced them within local school systems so that job training and certification are synchronized with emerging workforce priorities. By creating and promoting viable career pathways with local industry partners, these innovation centers both surface and meet emerging trends, demonstrating a value-add to all parties. As with all facets of a partnership, ensuring alignment across all stakeholders is key.

In addition to meeting labor market and regional economic development priorities, industry partners benefit directly from a healthy entrepreneurial community within an innovation center. Strong UI partnerships often function as an expanded base for research and development because they have established and respected processes for the protection of intellectual property.

Kendall Square, located in Cambridge, MA, is a clear example of the impact of intentional networking among a group of academic and industry partners. Within Kendall Square, the Cambridge Innovation Center (CIC) launched in 1999 as an independent organization to promote coworking and aligned research development across a number of nearby universities and research centers. Kendall Square also serves as a coordinated entrepreneurship ecosystem, promoting early stage development and investment by creating a social and professional hub of activity. The CIC and research park have both grown together over the past three decades. According to the CIC website, there are now over 10,000 CIC client organizations. The CIC hosts over 4000 events globally, and CIC clients have attracted over \$17 billion of investment. These efforts simultaneously benefit the greater-Boston community and the academic community at each individual institution. Most importantly, perhaps, they appear to have spurred a self-reinforcing cycle of investment and co-location among industry and entrepreneurial partners.

7.3.2 *Academic Presence*

Another precondition for a successful UI partnership is, unsurprisingly, a strong academic presence. As previously noted, universities have incredible potential to serve as a fertile ground upon which a research park can be developed. That said, universities demonstrate their commitment to collaborative research environments

through serious, top-level leadership to these endeavors and by leveraging physical and intangible infrastructure that supports its growth. Ultimately, research parks may struggle to successfully leverage the full potential of new or revitalized physical assets without internal champions or adequate university investment.

Internal champions are critical to navigating challenges regarding physical space. For example, Georgia Tech's then-President Wayne Clough has been referred to as "The Father of Tech Square" for his persistence in securing public and private funding, as well as his willingness to make difficult choices to create a centralized hub of activity around Tech Square when it was initially conceived in the late 1990s. Clough's book, *The Technological University Reimagined, Georgia Institute of Technology, 1994–2008* provides concrete examples of the extent to which Georgia Tech's transformation came about through strategic vision and leadership.

To extrapolate to other potential UI collaborations, a few key takeaways emerge. Namely, the university was led by a strong internal champion who negotiated a variety of physical and organizational hurdles in order to create a new physical footprint for the university, understanding that Tech Square's long-term vibrancy and success would also support the broader university and community.

Additionally, in partnering to develop Tech Square in a formerly undesirable and physically removed footprint, the Georgia Tech Foundation agreed to invest new funding provided that the university also made a commitment to relocate some of its academic enterprise in the newly developed Tech Square. Ultimately, Tech Square was anchored by Georgia Tech's Management School. Given the university requirement that every Georgia Tech student take at least one management class during their undergraduate course of study, Tech Square naturally became a familiar, frequented location and a hub of activity. Leadership-level choices about the university's programming and the physical location of a research park are pivotal to ensuring an innovation center's relevance and long-term viability.

When research parks can align existing physical and intangible assets, they also create a more coherent case for support from both public and private partners. Given that all UI collaborations will require early investment to help seed and launch an innovation center, the efficient use of existing personnel and infrastructure is all the more important. Furthermore, aligning existing resources will make initial fundraising, messaging, and community engagement much easier to translate to prospective partners, both within industry and among relevant regional stakeholders.

Ensuring alignment across diffuse and often siloed university departments is another important component of academic leadership in UI partnerships. To help keep as many different stakeholders clear on the overall vision as possible, UI partnerships require academic institutions to demonstrate and articulate a clear value proposition. By having a clear vision of its role and processes, academic partners ensure consistency in communication and make it far easier to attract external partners. Clarity from academic leadership and alignment across the university also avoids duplicative conversations or redundancies, ensuring that administrative or bureaucratic processes don't cause unnecessary obstacles to a research park's development.

Perhaps one of the most prominent examples of a UI collaboration that leveraged existing academic capacity towards improved economic development efforts is Research Triangle Park (RTP), which brought together Duke University, the University of North Carolina, Chapel Hill, and North Carolina State University—and their three distinct geographic areas—to create a unified organization that would benefit all stakeholders.

Originally, RTP began as an improbable concept: a for-profit venture spearheaded by the president of a regional bank and a local developer, with support from then-governor Luther Hodges, as a way to help reverse the state's economic decline and stem the departure of its graduates to more prosperous regions. The original vision for RTP could not have anticipated the growth and success of the ensuing decades, and there were few, if any, examples of a regional hub that leveraged academic infrastructure to formally align with industry partners and develop transferable technology.

To surmount initial skepticism from academic, industry, and local government partners, RTP was intentional about promoting collaboration and density through its programming and its physical assets. Furthermore, RTP developed programming that explicitly responded to the needs of its state and regional community. Eventually, with support from an array of investors, including individual citizens of North Carolina and institutional partners, RTP also attracted support from the state legislature of North Carolina and was also able to secure prominent industry investment. In fact, an initial anchor investment came from IBM, which agreed to build a 600,000 square foot facility on the campus. By this point, RTP had begun to build a reputation as a strong resource for companies seeking to develop new technology, scale their research and development activities, and leverage underutilized talent from three different R1 universities.

While no single decision can be attributed to RTP's long-term success, RTP's President and CEO, Scott Levitan emphasized the importance of programming and placemaking that builds on existing infrastructure and talents in a particular region. Furthermore, Mr. Levitan pointed out that RTP, in its early stages, made decisions that encouraged increased investment, which then further encouraged industry participation, creating a virtuous cycle of development.³

In practice today, RTP appears to have become more than an agglomeration of buildings and activities. Like any successful UI collaboration, it has become its own vibrant community, an R&D center with placemaking and programming that can adapt to dynamic regional economic and labor-market demands. Today, for example, RTP boasts a variety of mixed-use venues that attract residents and visitors of all types, with plans in place to develop additional mixed-use buildings and infrastructure that will encourage meaningful interactions among entrepreneurs, academics, and industry partners.

³Mr. Levitan was formerly the Vice President of Real Estate at the Georgia Institute of Technology during the development of Tech Square. He later moved to Forest City Realty Trust before ultimately assuming his current role as CEO of RTP.

Challenges can arise when universities face physical limitations to the creation of a new research park or when they lack a coherent vision for their endeavor. In some cases, a university might mistakenly believe that a research park alone can generate a community of innovation by attempting to capture a trend from outside the region or through a single external partner. Instead of falling into this trap, academic institutions can take an incremental approach to the development of a research park. Some universities have built centers where continuing education activities, like certificate or training centers in satellite locations, become a value-add to an existing academic and industry community without attempting to create an entirely new community of practice where none currently exists.

7.3.3 Academic and Public Commitment to R&D

Given that the foundation of a research park is academic research, strong and mature research programs will attract additional collocation and build a virtuous cycle of entrepreneurship and UI partnerships. Universities can demonstrate their commitment to research and development through programming that incentivizes activities that are of mutual benefit to the institution and its current or potential industry partners.

Purdue University's Discovery Park provides a strong example of the power of shared incentives and also underscores the potential impact of exploiting underutilized real estate to attract and retain talent. Discovery Park began with an initial investment of just over \$30 million from both the Lilly Endowment and the state of Indiana to create an expanded physical presence in the city of West Lafayette, which sits on the western border of Purdue University. Initially, industry partners were focused on agricultural, chemical, and aerospace engineering and attracted major partnerships with Bayer, Rolls-Royce, and Saab.

Today, Discovery Park has been transformed into Discovery District and the initial \$30 million investment has generated over one billion dollars of funding and research within a self-sustaining enterprise. The Purdue Discovery District now features research centers and institutes representing broader interests like nanotechnology and advanced computational studies within a group of proximate buildings. The Discovery District website prominently features the success of its research and development efforts at the same time as it underscores the comprehensive value to prospective residents, featuring the area's walkability, top-tier public schools, and rapidly expanding mixed-use real estate and infrastructure in West Lafayette. Discovery District is now one of the largest university-affiliated research parks in the country, having developed underutilized physical space and created opportunities for Purdue's talented faculty and students to participate in a coordinated ecosystem of academic and entrepreneurial activities.

As outlined in the example above, in many cases a university's initial investments in a new research enterprise can come through leveraging existing resources, including physical assets like partially vacant buildings or underutilized real estate

parcels. Additionally, universities can leverage available personnel to serve as savvy translators and relationship brokers within state or university bureaucracies. University administrators and staff may serve as important partners by supporting the use of state and federal funds, either by creating room within existing grant budgets or allocating space in academic buildings to serve as a pilot study of larger potential collaborations.

There are some relatively common obstacles to securing support for a UI partnership. For example, many small hurdles may arise related to physical space and relational dynamics within a large university bureaucracy. When departments or university-affiliated institutes do not have a common goal or an incentive to partner, communication gaps or conflicts may result.

One of the most obvious remedies to these challenges can come through strong leadership and in the coordinated physical presence of an academic partner. A clear and compelling vision from university leadership, as well as strong processes for communication and coordination across the university will also help reassure established industry partners and nascent entrepreneurs that their participation in an innovation center will be protected and encouraged. When institutions signal and demonstrate a strong commitment to external transfer, it can create a virtuous cycle of activity in the park.

Universities have also enacted specific policies to incentivize academic participation in research parks. As a result, the overall academic posture towards industry partnerships has changed fairly significantly in the past few decades. For example, many universities have now made entrepreneurship or commitment to industry-affiliated research an explicit part of the promotion and tenure process, which has dramatically changed the perception of UI collaboratives among academics.

When universities are able to leverage initial partnerships, the virtuous cycle can be quite impressive. For example, Central Florida Research Park was founded in 1978 through a partnership with the University of Central Florida. The Naval Air Warfare Center Training Systems Division relocated to the park in 1993. Since that relocation, the university and state have invested to attract additional defense department tenants to the park and it now has over 9500 employees and tenants including Northrop Grumman, Boeing, Leidos, ManTech, General Dynamics, Parsons, and Booz Allen Hamilton. UCF's R&D expenditures have more than doubled in the last decade from \$109M in 2011 to \$239M in 2020.

7.3.4 Fundraising and External Relations

Fundraising, messaging, and holistic visioning are central to the success of a research park. There are numerous examples of university leaders whose efforts spurred public-private partnerships that led to lasting economic development. However, these initiatives did not materialize unaided or from a singular visionary. Instead, they were the product of a preexisting community of practice and were almost always the result of careful planning and stakeholder engagement.

For example, initial funding for Research Triangle Park came in part from what is known today as a “roadshow”—a statewide fundraising effort—that ultimately raised over \$1.5M in 1958 (\$15M in current dollars). Even though the park was concentrated in one region, North Carolinians from all parts of the state and all walks of life understood the value of transforming the state’s economy and reversing the “brain drain” and were therefore more incentivized to participate. This type of coordinated and diverse fundraising remains important to the success of new and ongoing research parks, as President and CEO Scott Levitan noted in conversations with authors of this chapter.

State and federal support for research parks remains an important support mechanism for research parks. For example, even though Massachusetts is home to the highest-ranked life sciences cluster in the country, the state still maintains a \$1.6 billion fund to support workforce training, tax and infrastructure development credits. The fund also seeds capital grants to support development and commercialization activities throughout the state. Similarly, NCInnovation is a \$500 million statewide endowment to spur partnership and successful commercialization throughout the state’s university systems.

Successful parks also adhere to the self-professed values of innovation and collaboration in their marketing and fundraising plans. By remaining flexible and leveraging existing sources of capital from within the university, regional economic development agencies, and state/federal grants, universities have been able to unlock layers of additional financing to accelerate their growth. Faculty at the University of Delaware were instrumental in securing a \$70 million National Institute of Standards and Technology grant in 2016 to establish The National Institute for Innovation in Manufacturing Biopharmaceuticals. In 2020, a \$165 million, state-of-the-art facility opened in their STAR Park.

Other parks and centers have deliberately created fundraising mechanisms that leverage existing regional infrastructure or take advantage of organic marketing or place-based incentives. Philadelphia’s University City Science Center website is targeted at students, researchers, startups, corporations, investors, and neighbors. Content is posted daily across all social media channels, and both a monthly newsletter and annual report are published. Integrated and comprehensive fundraising and communication strategies will more effectively engage the largest number of stakeholders.

To achieve their fundraising goals, research parks should pay careful attention to ongoing state and federal funding opportunities, which have increased dramatically in the past two decades. The federal funding landscape includes relatively new opportunities through both the United States Economic Development Agency (EDA) and the National Science Foundation (NSF). In 2022, over \$10 billion was appropriated for up to five years through the CHIPS (Creating Helpful Incentives to Produce Semiconductors) Act to create a “Tech Hubs” program across the country. In 2023, \$500 million was allocated to officially launch the program and in February 2024, 12 locations were selected to receive capacity-building grants to cover scaling emergent technologies.

Through its Technology, Innovation, and Partnerships (TIP) Directorate, the NSF also received federal funding from the CHIPS and Science Act of 2022. In January 2024, they awarded ten regional innovation collaboratives with up to \$160 million of funding over ten years after awarding 44 different capacity-building grants to teams across the country.

Taken all together, these pieces of major federal legislation and their subsequent implementation are a clear sign of strong bipartisan support for innovation centers, as technology development and transfer are now recognized as drivers of economic growth and mobility. The implication for aspiring innovation centers is to maintain an awareness of ongoing and special opportunities from state and federal sources, focusing on the appropriate agencies and individual stakeholders administering these funding opportunities.

Fundraising can be complicated and uncertain, however, and challenges are not uncommon. Often, unaffiliated research parks will need to carefully consider how to generate revenue to pay for their administrative and management activities. One creative example is the Sandia Science and Technology Park in Albuquerque, NM, which generates revenue by providing fiber data and communication services to tenants and partners. In other cases, there are funding considerations among industry partners or within a larger university funding picture that can influence competitiveness for state and federal grants. Other times, if a university does not serve as a public champion or does not broadcast a consistent message to external audiences, it can lead to inconsistent awareness or lackluster fundraising. To combat these challenges, research institutions can leverage their considerable personnel and physical assets to create a strong scaffolding on which to develop a new ecosystem, without needing to attract as much initial seed capital.

7.3.5 *Regional Collaboration*

Regional collaboration refers to the coordination required among disparate stakeholders to collaboratively build a community of practitioners. Effective collaboration will almost certainly require different participants and strategies at different points in a park's lifecycle. That said, collaboration will always require initial staff members who are on the ground and can spearhead the visioning, external relations, and fundraising efforts necessary to scale. It is particularly important for the executive to serve as a project manager who can truly live and breathe the daily activities of a research park. There is no one requisite resume or skill set, but ideally an executive will be a multifaceted individual who brings or can quickly build skills in real estate, government relations, finance, marketing, academia, program activation, and property management. Different research parks will have different business plans and therefore unique strategies and operations. Therefore, creating a clear initial scope of work will be pivotal in setting up the first park administrator for success.

Regional collaboration benefits from an intentional networking effort should aim to take advantage of regional and state economic development agencies, university

fundraising and corporate engagement staff, and commercial partners. For example, at the University of Illinois Research Park, the executive director also oversees the University's Economic Development Office and has full-time positions on staff that include a Director of External Engagement, a Director of EnterpriseWorks (an incubator), a Corporate Events and Program Director, and an Outreach Coordinator for Innovation and Entrepreneurship.

In this example, the executive has direct exposure to industry partners working with scientists at the school and, simultaneously, university advancement and external relations teams. When the executive is connected personally to external partners and internal departments, they will be better able to anticipate their individual priorities and their areas of potential overlap.

As with any collaborative effort, challenges can occur with poor communication and coordination, specifically when interpersonal relationships or divergent professional priorities can stifle the well-being of the group at large. Innovation centers can help mitigate these obstacles by creating mutually dependent systems for communication and coordination. These systems should prioritize transparency and agility, ensuring communication is abundant and free flowing.

Georgia's Stanton Springs Joint Development Authority is a model for cooperation between four counties, whose site attracted a major pharmaceutical factory, a state-funded workforce development center, and an electric vehicle manufacturing plant, creating 9000 new jobs and \$6 billion in total investment. These efforts were supported by the state university system and also featured extensive involvement of the technical college system for job training. In this example, a state government helped achieve its priorities for job training and economic development through both state universities and technical college systems, requiring equal participation and collaboration from top academic researchers and the workforce development partners in the technical colleges to train and upskill a labor force.

New funding opportunities from state and federal sources appear likely to continue to facilitate these sorts of regional collaborations across multiple universities and broader geographies. As noted above, the EDA and the NSF TIP Directorate are mandated to facilitate regional hubs for development and commercialization that expand beyond single anchor institutions and instead facilitate regional, multi-sector collaboration.

UI partnerships must also take into account the complications that can arise when attempting to simultaneously balance university and regional bureaucracies. In addition to asset mapping, personal relationships within these systems can help avoid gridlock and promote collaboration. Universities and state university systems can more systematically support partnerships through clear guidance and administrative support that balance entrepreneurs', industry partners', and university researchers' varying priorities, all while creating clear protections for intellectual property. The best solutions will manage these dynamics and lead to the improvement of a community as a whole.

Regional collaboration cannot be preordained or pre-programmed: it needs to be native to a particular place and should intend to capitalize on a region's unique cultural and geographic characteristics. In this way, innovation centers should be

scaffolded on the organic assets of a community. In particular, they will take into account cultural considerations regarding the quality of life and design activities in an innovation center that emphasizes these strengths.

7.3.6 Activation, Events, and Placemaking

Activation and events that promote culture and community building should be carefully considered when creating successful UI partnerships that seek to promote long-lasting economic development.

Creating organic and responsive activities that spark enthusiastic participation and engagement will require clear communication among research park staff and its various stakeholders. TUFF has learned that taking great pains to collect data during events and frequently asking questions will equip an innovation center to continuously improve its programming and avoid stagnancy. TUFF considers activation and events as creating value “in” the building and not “of” the building—reinforcing the idea that physical assets on their own will not be the impetus for lasting economic growth and development.

Placemaking events will naturally combine the interests and capacities of a research park’s diverse stakeholders and will leverage the unique cultural and demographic characteristics of a particular region. When these core activities are thus aligned, they will nurture a community with strong bonds and a self-reinforcing culture. For some building managers, this may mean shifting from what has traditionally been a more passive form of property management to a more active and community-oriented approach. In so doing, innovation centers will create a culture of citizenship and participation that is, by definition, concerned with the well-being of the whole community.

At Tech Square, TUFF’s Tom Hall worked hard to innovate this concept of unconventional or unique activation to get more organic collaboration among very serious-minded individuals who might not naturally have gravitated toward socialization. In TUFF’s experience, the type of event should be very flexible and gather as many people together as possible: Lego building contests, trivia events, board games, taco Tuesdays—anything to get people talking to each other and forming real bonds. TUFF has a motto that “density is destiny,” which refers not only to physical proximity but also to the frequency and quality of the human interactions promoted by strong, responsive programming. If, as is the case for most innovation centers, the ultimate goal is economic growth and development, innovation centers should create environments of trust. In this way, TUFF considers commerce as enabled by community, and community enabled by the density of human experience and lasting interpersonal bonds.

It can be challenging to create responsive programming and to build culture among a widely disparate group of individuals, especially among the scientific community. It can also be challenging to collect and measure data for placemaking events, or to build a mechanism for continuous feedback and improvement.

Innovation centers can counteract these challenges through intentional decisions regarding its people and activities. Specifically, staff members need to have the explicit purview to build relationships, ask questions, and create events that meet tenant interests. Relatedly, by creating processes and tools that then enable a UI effort to measure engagement and interest among tenants, they will have a running knowledge of the most successful activities. Knowing that cultures shift as industries and regions evolve, innovation centers must be ready to adapt and change. By taking a long-term approach to the physical assets and considering themselves partners to a region's economic well-being, UI partnerships will be best positioned for success.

7.4 Summary

There is a clear link between federal funding for UI partnerships and the growth of the modern day innovation district. Most recently, the Inflation Reduction Act, the Build Back Better Act, and the CHIPS and Science Act collectively authorized hundreds of billions in funding and incentives for innovation districts and the development of scientific technology. These federal policies, in concert with a maturing venture capital presence in and around universities, has spurred increased awareness of the potential for commercialization and technology transfer. As noted in a relatively recent analysis by the Brookings Institution, a large percentage of new federal spending allocated towards innovation districts will be focused on areas of the country with less-established infrastructure and economies (Hausman, 2012). As a result, aspiring UI collaborations now face both a more competitive funding environment and an evolving landscape in which to demonstrate their capacity to manage a successful collaboration.

As outlined previously, both the NSF and EDA have ongoing and robust funding opportunities that appear likely to lead to new rounds of funding and incentives in the coming years. As a short-term strategy, UI collaboratives should familiarize themselves with the individuals and agencies responsible for funding allocation and implementation.

UI collaborations will almost certainly need to first succeed in the planning and stakeholder engagement activities outlined previously in this chapter, but it is important to recognize that ideas and processes will naturally evolve. When the ultimate goal is economic development, there are often unexpected and unanticipated routes to shared success and aligned interests. With advice and partnership from a network of carefully cultivated partners and experts, UI collaboratives will be better positioned to more efficiently uncover and leverage their regional strengths.

Additionally, given the varying levels of existing industry or academic infrastructure in some parts of the country poised for an influx of investment, new or unanticipated obstacles will likely arise. These challenges will, by definition, be different from those among more established economies, larger cities, or "tier 1" research institutions. For example, when facing inevitable market fluctuations,

legislative obstacles, or decreasing commercial viability in a particular technology, smaller regions will need to consider how to build resilient systems. Emerging technologies like artificial intelligence and other low-code systems should make it easier for UI collaborations to collect and evaluate data on their relative progress toward goals. It would be wise for newer or less resourced UI partnerships to consider these tools as a means to garner increased funding for less than was previously possible.

Given the multitude of factors that UI collaborations have to consider, TUFF recommends considering “five steps” for moving beyond ideation or planning:

1. **Act:** Move past planning alone; begin to generate tangible progress through activities.
2. **Iterate:** The process will continue to improve your initial concept.
3. **Network:** Talk to as many different people as possible to draw inspiration and make unexpected connections.
4. **Persist:** Pursue opportunities that you deem to be a fit even if there is initial reluctance.
5. **Seek:** Look for advice and guidance from experts so you do not reinvent what you are already doing.

New or developing industries like artificial intelligence, human-machine interfaces, advanced manufacturing, and quantum information technologies are among the areas most prioritized in new federal funding. There is increasingly more research in these fields but there are fewer established innovation centers supporting them. This may create opportunities for both incumbent and nascent collaboratives, but they will require nimble and responsive approaches. In some cases, collaboratives that attempt to be too responsive to shifting landscapes might risk distorting their core value proposition and negatively affecting their long-term impact.

7.5 Takeaways and Future Strategies

Successful innovation centers can significantly influence the trajectory of a research institution and its regional economy. In this chapter, TUFF has endeavored to provide actionable advice and specific examples garnered from over four decades of direct experience in this space. Specifically, our aim is to articulate that successful innovation centers should view their overarching goal as increasing the standard of living in a given community. Using this philosophy and approach, TUFF believes that UI collaborations can better achieve the disparate aims among a wide variety of stakeholders involved in any UI collaboration.

TUFF’s framework has four core tenets for a successful innovation community:

- High-quality academic institutions
- Private investment in research and development
- Collaboration between university and industry
- Protection of IP

To support these four pillars, TUFF has outlined six priority areas that UI collaborations should carefully consider:

1. Academic presence
2. Academic and public support, commitment to R&D
3. Fundraising and external relations
4. Regional collaboration
5. Entrepreneurship and asset mapping
6. Activation and events, placemaking in real estate

Certainly, however, there are myriad additional factors at play and differences in implementation due to inherent differences across geographies. To state it plainly, there is simply no one-size-fits-all approach. Our experience has shown four areas of success:

1. Tech-Square is an ideal and ever-changing lens through which to view the relative strength and resilience of UI collaborations and their long-term impact on a regional economy. Tech-Square has weathered two of our nation's most severe recessions and a global pandemic that disrupted both the commercial real estate market and irrevocably altered the psyche of the American worker—seemingly permanently. TUFF believes that innovation centers will be resilient through successful culture-building, long-term investment, and truly responsive placemaking in real estate.
2. On a related note, difficulties now facing the commercial real estate market and brought on by the Covid-19 pandemic seem to emphasize the importance of many UI fundamentals outlined in this chapter. In this way, commercial real estate should not be considered equivalent to nuanced UI development as outlined in this chapter. Innovation districts that feature strong UI partnerships have a distinct advantage in developing physical and intangible assets that are mutually reinforcing and therefore resilient to evolving market conditions.
3. Creative placemaking, relationship development, and fundamental partnership building is as important as ever. There are no guaranteed successes in UI partnerships, but collaborations will be more successful when they seek to align with a community's existing infrastructure and assets. When aligning a new job training center in a smaller, less developed region, for example, a research park with a nascent UI collaboration should pay careful attention to the history and expertise of the state and design around those as much as possible.
4. Partnerships should build a concrete awareness of their regional differentiators and then leverage those assets to build from a strong core. By remaining true to the unique people and characteristics of a given place, all of the aspects of a collaborative should be easier, from fundraising and marketing through to the complicated coordination across public and private stakeholders.

References

- Battelle Technology Partnership Practice. (2013). *Driving regional innovation and growth: Results from the 2012*. Survey of North American University Research Park.
- Chai, S., & Shih, W. (2016). Bridging science and technology through academic–industry partnerships, *Research Policy, Elsevier*, 45(1), 148–158.
- Cullum Clark, J. H. (2022). *Immigrants and opportunity America's cities* (Blueprint for opportunity series No. 3). The George W. Bush Institute -SMU Economic Growth Initiative.
- Darmody, B., & Bendis, R. (2021). Creating communities of life science innovation in the US: History of critical factors that helped the BioHealth capital region emerge. *Association of University Research Parks*, 26(1), 27–39.
- Dooley, L., & Kirk, D. (2007). University-industry collaboration: Grafting the entrepreneurial paradigm onto academic structures. *European Journal of Innovation Management*, 10(3), 316–332.
- Hausman, N. (2012). *University innovation, local economic growth, and entrepreneurship*. Harvard University Center of Economic Studies. www2.census.gov/ces/wp/2012/CES-WP-12-10.pdf
- Hutton, T. (2004). The new economy of the inner city. *Cities*, 21(2), 89–108.
- Katz, M., & Ordover, J. (1990). R&D cooperation and competition. *Brookings Papers on Economic Activity, Economic Studies Program*, 21, 137–203.
- Rissola, G., Bevilacqua, C., Monardo, B., & Trillo, C. (2019). *Boston-Cambridge innovation districts — USA*. European Research Center. <https://ec.europa.eu/jrc>

Kevin T. Byrne is the President and CEO of TUFF, recognized for his innovative solutions to capital challenges faced by institutions. He specializes in non-traditional public–private partnerships and real estate financing for university research parks and innovation districts. Byrne holds an MBA with honors from the Wharton School and graduated with honors from Furman University, where he now serves as Chairman of the Board of Trustees. He has served as President of the Association of University Research Parks and is involved in various community and advisory boards, including Young Life Metro Atlanta and the Boy Scouts of America.

Victor R. Clements is a seasoned CPA with over 35 years of experience in public accounting and senior finance positions, specializing in financing strategies and environmentally sustainable building solutions. A 1987 graduate of Georgia Southern University, Clements has managed over \$2 billion in capital market transactions, developed sustainability initiatives saving over 38%, and ensured compliance with tax and audit requirements for a variety of organizations. At TUFF, Clements oversees all operational activities, asset management and financing strategies. He is a member of multiple professional organizations and actively supports community initiatives, including Southface Institute and the Atlanta Children's Shelter.